Partial Differential Equations

Lagrange's Linear PDE

Solution of Lagrange's Linear PDE

The general solution of the linear PDE

$$Pp + Qq = R$$

is

$$\phi(u, v) = 0,$$

where ϕ is an arbitrary function, and

$$u(x, y, z) = c_1, \qquad v(x, y, z) = c_2$$

form a solution of the equations

$$\boxed{\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}},$$

which are called **subsidiary equations**.

Example 1: Find the general integral of $xp + yq = z$

Step 1. Compare

$$Pp + Qq = R \qquad \text{and} \qquad xp + yq = z.$$

We get

$$P = x, \qquad Q = y, \qquad R = z.$$

Step 2. Form the auxiliary simultaneous equations:

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R} \quad \Longrightarrow \quad \boxed{\frac{dx}{x} = \frac{dy}{y} = \frac{dz}{z}}.$$

Solving the auxiliary equations

Solve

$$\frac{dx}{x} = \frac{dy}{y}, \qquad \frac{dy}{y} = \frac{dz}{z}.$$

For the first pair,

$$\int \frac{dx}{x} = \int \frac{dy}{y} \Longrightarrow \ln|x| = \ln|y| + C_1,$$

so

$$\ln|x| - \ln|y| = C_1.$$

Exponentiating both sides gives

$$\left|\frac{x}{y}\right| = e^{C_1}.$$

Since the right-hand side is a constant, we write

$$\boxed{\frac{x}{y} = c_1}.$$

Second independent solution

Similarly,

$$\int \frac{dy}{y} = \int \frac{dz}{z} \Longrightarrow \ln|y| = \ln|z| + C_2,$$

so

$$\ln|y| - \ln|z| = C_2.$$

Exponentiating both sides gives

$$\left|\frac{y}{z}\right| = e^{C_2},$$

and hence

$$\boxed{\frac{y}{z} = c_2}.$$

Thus two independent solutions are

$$u(x,y,z) = \frac{x}{y} = c_1, \qquad v(x,y,z) = \frac{y}{z} = c_2.$$

General integral

The solution is therefore written as

$$\phi(u, v) = 0.$$

Substituting the two independent integrals gives

$$\boxed{\phi\left(\frac{x}{y}, \frac{y}{z}\right) = 0}.$$

Method of Multipliers

This method helps us to solve subsidiary equations. Starting from

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R},$$

introduce a common parameter ds:

$$\boxed{\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R} = ds}.$$

Thus

$$dx = P\,ds, \qquad dy = Q\,ds, \qquad dz = R\,ds.$$

Let l, m, n be multipliers. Multiplying these equations respectively by l, m, n and adding gives

$$l\,dx + m\,dy + n\,dz = (lP + mQ + nR)ds.$$

Method of Multipliers: Main formula

Provided $lP + mQ + nR \neq 0$,

$$ds = \frac{l\,dx + m\,dy + n\,dz}{lP + mQ + nR}.$$

Therefore,

$$\boxed{\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R} = \frac{l\,dx + m\,dy + n\,dz}{lP + mQ + nR}}.$$

The multipliers l, m, n may be constants or suitably chosen functions of x, y, z.

Method of Multipliers: Obtaining first integrals

Choose l, m, n such that

$$\boxed{lP + mQ + nR = 0}.$$

Then

$$l\,dx + m\,dy + n\,dz = 0.$$

If this is a perfect differential of some function, say $u(x, y, z)$, then

$$du = 0.$$

Integrating gives

$$\boxed{u = c_1}.$$

Let us explain why. Suppose there exists a differentiable function

$$u = u(x, y, z)$$

whose total differential is

$$du = \frac{\partial u}{\partial x}dx + \frac{\partial u}{\partial y}dy + \frac{\partial u}{\partial z}dz.$$

For simplicity, write

$$du = u_x\,dx + u_y\,dy + u_z\,dz.$$

If we can choose the multipliers such that

$$l = u_x, \qquad m = u_y, \qquad n = u_z,$$

then

$$l\,dx + m\,dy + n\,dz = du.$$

But from equation (2),

$$l\,dx + m\,dy + n\,dz = 0.$$

Therefore,

$$\boxed{du = 0.}$$

Integrating both sides gives

$$\int du = \int 0.$$

Thus,

$$\boxed{u(x, y, z) = C_1.} \tag{3}$$

This is called the first integral of the auxiliary equations.
Similarly, choose another set of multipliers l', m', n' such that

$$l'P + m'Q + n'R = 0.$$

If the corresponding expression is integrable, it gives a second integral

$$\boxed{v = c_2}.$$

Hence the general solution is

$$\boxed{\phi(u, v) = 0 \quad \text{or} \quad u = f(v)},$$

where the function is arbitrary.

Since the PDE does not prescribe a particular relationship between the two constants, we can relate them using an arbitrary function. For example, suppose

$$c_1 = f(c_2).$$

Substituting our first and second integrals gives

$$u = f(v).$$

Moving everything to the left-hand side,

$$u - f(v) = 0.$$

Now define a new function

$$\phi(u, v) = u - f(v).$$

Therefore,

$$\boxed{\phi(u, v) = 0.}$$

Procedure for Solving $Pp + Qq = R$

To solve the equation

$$\boxed{Pp + Qq = R}:$$

1. Form the auxiliary simultaneous equations

$$\boxed{\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}}.$$

2. Solve these equations to obtain two independent solutions

$$u = c_1, \qquad v = c_2.$$

3. Write the general solution as

$$\boxed{\phi(u, v) = 0 \quad \text{or} \quad u = f(v)},$$

where f is arbitrary.

Example 2: A Lagrange equation

Use the method of multipliers to find the general solution of

$$x(z^2 - y^2)p + y(x^2 - z^2)q = z(y^2 - x^2).$$

Answer: Comparing with $Pp + Qq = R$ gives

$$P = x(z^2 - y^2), \qquad Q = y(x^2 - z^2), \qquad R = z(y^2 - x^2).$$

The subsidiary equations are

$$\boxed{\frac{dx}{x(z^2 - y^2)} = \frac{dy}{y(x^2 - z^2)} = \frac{dz}{z(y^2 - x^2)}}.$$

First set of multipliers

We know that

$$lP + mQ + nR = lx(z^2 - y^2) + my(x^2 - z^2) + nz(y^2 - x^2) = 0.$$

, what can we choose as l, m and n? We have to take the multipliers x, y, z so that

$$xP + yQ + zR = x^2(z^2 - y^2) + y^2(x^2 - z^2) + z^2(y^2 - x^2) = 0.$$

Hence

$$l\,dx + m\,dy + n\,dz = 0$$

and thus

$$x\,dx + y\,dy + z\,dz = 0.$$

Integrating,

$$\int x\,dx + \int y\,dy + \int z\,dz = C,$$

so

$$\frac{x^2}{2} + \frac{y^2}{2} + \frac{z^2}{2} = C.$$

Thus the first integral may be written as

$$\boxed{u = x^2 + y^2 + z^2 = C_1}.$$

Second set of multipliers cont...

To find the second integral take the multipliers

$$\frac{1}{x}, \qquad \frac{1}{y}, \qquad \frac{1}{z}.$$

Then

$$\frac{P}{x} + \frac{Q}{y} + \frac{R}{z} = (z^2 - y^2) + (x^2 - z^2) + (y^2 - x^2) = 0.$$

Hence

$$\frac{dx}{x} + \frac{dy}{y} + \frac{dz}{z} = 0.$$

Integrating,

$$\ln x + \ln y + \ln z = C_2,$$

and therefore

$$\boxed{v = xyz = C_3}.$$

Thus the general solution is

$$\boxed{\phi(u, v) = \phi(x^2 + y^2 + z^2, xyz) = 0}.$$

General Form of Second-Order PDEs

A general linear second-order PDE in two independent variables can be written as

$$\boxed{Au_{xx} + 2Bu_{xy} + Cu_{yy} + Du_x + Eu_y + Fu = f}.$$

If

$$f = 0,$$

the equation is **homogeneous**. If

$$f \neq 0,$$

the equation is **non-homogeneous**.
The coefficients A, B, C, D, E, F may be constants or functions of x and y, and f is a function of x and y.

Examples of Linear PDEs

Examples recorded in the notes include equations of the forms

$$(x + 2y)u_x + u_y = \sin(x^2 + y^2),$$

which is a non-homogeneous linear PDE of first order, and

$$xu_{xx} + yu_{yy} + u_x = \sin x,$$

which is a non-homogeneous linear PDE of second order.

Classification of Second-Order PDEs

Consider

$$Au_{xx} + 2Bu_{xy} + Cu_{yy} + Du_x + Eu_y + Fu = 0.$$

Associate with the second-order part the quadratic expression

$$P(\alpha, \beta) = A\alpha^2 + 2B\alpha\beta + C\beta^2.$$

Its discriminant is

$$\boxed{B^2 - AC}.$$

Therefore:

$$B^2 - AC > 0 \quad \Rightarrow \quad \text{hyperbolic},$$

$$B^2 - AC = 0 \quad \Rightarrow \quad \text{parabolic},$$

$$B^2 - AC < 0 \quad \Rightarrow \quad \text{elliptic}.$$

Homogeneous Constant-Coefficient PDEs

Define

$$D_x = \frac{\partial}{\partial x}, \qquad D_y = \frac{\partial}{\partial y}.$$

We seek solutions of an equation of the type

$$\boxed{\frac{\partial^2 u}{\partial x^2} + k_1 \frac{\partial^2 u}{\partial x \partial y} + k_2 \frac{\partial^2 u}{\partial y^2} = 0},$$

where k_1 and k_2 are constants. Equivalently,

$$\boxed{\left(D_x^2 + k_1 D_x D_y + k_2 D_y^2\right) u = 0}.$$

Auxiliary Equation

The auxiliary equation is obtained by replacing

$$D_x \longrightarrow m, \qquad D_y \longrightarrow 1.$$

Thus

$$\boxed{m^2 + k_1 m + k_2 = 0}.$$

For distinct roots, solving this equation gives

$$m = m_1, \qquad m = m_2.$$

These values allow us to factorise the original PDE as

$$\boxed{(D_x - m_1 D_y)(D_x - m_2 D_y)u = 0}.$$

This is the factorised form of the second-order PDE.

Solving the First Factor

First solve

$$(D_x - m_2 D_y)u = 0.$$

This means

$$\frac{\partial u}{\partial x} - m_2 \frac{\partial u}{\partial y} = 0.$$

Writing

$$p = \frac{\partial u}{\partial x}, \qquad q = \frac{\partial u}{\partial y},$$

the equation becomes

$$\boxed{p - m_2 q = 0}.$$

This is a first-order linear PDE.

Characteristic System for the First Factor

Recall that a first-order PDE

$$Pp + Qq = R$$

has the characteristic auxiliary system

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{du}{R}.$$

For

$$p - m_2 q = 0,$$

we have

$$P = 1, \qquad Q = -m_2, \qquad R = 0.$$

Therefore

$$\boxed{\frac{dx}{1} = \frac{dy}{-m_2} = \frac{du}{0}}.$$

First Factor: Characteristic Invariant

Solve

$$\frac{dx}{1} = \frac{dy}{-m_2}.$$

Then

$$-m_2\,dx = dy \quad \Longrightarrow \quad dy + m_2\,dx = 0.$$

Integrating,

$$\boxed{y + m_2 x = C}.$$

This means that along each characteristic curve, the quantity $y + m_2 x$ remains constant. The characteristic curves are

$$y = -m_2 x + C,$$

which are straight lines.

First Factor: General Solution

Now solve

$$\frac{dx}{1} = \frac{du}{0}.$$

This gives

$$du = 0 \quad \Longrightarrow \quad \int du = 0,$$

so

$$u = C_1.$$

The value of u need not be the same on every characteristic curve. Since each characteristic is identified by

$$C = y + m_2 x,$$

the constant C_1 may depend on C:

$$C_1 = \phi(C).$$

Therefore

$$\boxed{u_1(x, y) = \phi(y + m_2 x)}.$$

Solving the Second Factor

Finally solve

$$(D_x - m_1 D_y)u = 0.$$

Thus

$$\frac{\partial u}{\partial x} - m_1 \frac{\partial u}{\partial y} = 0,$$

or

$$p - m_1 q = 0.$$

Hence

$$P = 1, \qquad Q = -m_1, \qquad R = 0,$$

and the auxiliary system is

$$\boxed{\frac{dx}{1} = \frac{dy}{-m_1} = \frac{du}{0}}.$$

Second Factor and Complete Solution

From

$$\frac{dx}{1} = \frac{dy}{-m_1}$$

we get

$$dy + m_1 dx = 0,$$

and hence

$$\boxed{y + m_1 x = C_2}.$$

Also,

$$du = 0 \quad \Longrightarrow \quad u = C_3.$$

As before, C_3 may depend on C_2; write

$$C_3 = \psi(C_2).$$

Thus

$$\boxed{u_2(x, y) = \psi(y + m_1 x)}.$$

The complete solution is